

EduSphere 360

AI-Powered International School Management Platform

Bangalore, Karnataka India

PRODUCT REQUIREMENTS DOCUMENT V2.1

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Dev Model	Agile Sprint-Based, 6-Phase Delivery

This document has been prepared after extensive discovery, stakeholder interviews, curriculum analysis, and competitive research. It represents the single source of truth for the EduSphere 360 platform and must be treated as the foundation document for all engineering, design, and QA decisions.

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01 Project Vision and Strategic Objectives

This section captures the foundational reason behind EduSphere 360. Every design decision, every feature, and every API integration traces back to these intentions. The development team is encouraged to revisit this section whenever facing ambiguous product decisions.

1.1 The Problem We Are Solving

International schools in Bangalore and across India operate on a patchwork of legacy software, WhatsApp groups, printed registers, and disconnected spreadsheets. Here is what we kept hearing in our discovery interviews:

HEARD

We have 11 different logins for 11 different systems. It is exhausting. (School Operations Head)

I do not know how my child is doing until the quarterly report card. By then it is too late. (Parent, Grade 8)

I spend 3 hours a week filling in attendance sheets. I became a Clerk, not a teacher. (Class Teacher, IGCSE)

We have no way to know which teacher is struggling until a parent complaint comes in. (Vice Principal)

Our timetable is still done in Excel. The file is 8MB. It crashes every Monday. (Scheduling Coordinator)

These are not edge cases. These are systemic failures that EduSphere 360 is purpose-built to eliminate.

1.2 Vision Statement

Build a living, breathing digital nervous system for the modern international school where every student is seen, every teacher is supported, every parent is informed, and every administrator has the clarity to lead with confidence.

Saran, Product Strategy Lead, V-ONE Automation

1.3 Strategic Objectives

- Academic Excellence** **Critical** **AI Powered** Improve average student academic outcomes by 25% within the first academic year through AI-driven personalised interventions
- Teacher Empowerment** **Critical** **AI Powered** Reduce teacher administrative burden by 60% and increase instructional time through automation of recurring tasks
- Parent Transparency** **High** Achieve a Parent Satisfaction Score of 90%+ through real-time communication and progress visibility
- Operational Efficiency** **High** Consolidate 11+ disconnected tools into one unified platform, reducing operational overhead by 40%
- Data-Driven Leadership** **Critical** **AI Powered** Equip school leadership with predictive analytics to make proactive decisions instead of reactive ones
- Revenue Growth** **High** Improve fee collection efficiency by 35% and reduce default rates through intelligent reminders and payment plans
- Compliance and Safety** **Critical** Achieve 100% CBSE/IGCSE regulatory compliance and PDPB 2023 data protection standards

1.4 Success Metrics

Metric	Baseline	Target (12 months)	How Measured
Student academic improvement	No baseline data	+25% average grade improvement	AI Progress Report comparison
Teacher admin time savings	3 to 4 hours per week on admin	Under 45 min per week on admin	Teacher survey plus time-tracking log
Parent app daily active users	N/A, new product	Above 70% parent DAU	App analytics dashboard
Fee collection rate	72% on-time	Above 95% on-time	Finance module reporting
Incident response time	2 to 3 days manual	Under 4 hours	Incident management module
Data entry errors	15 to 20% error rate	Under 2% error rate	Audit log comparison
Staff NPS score	Unknown	NPS above 60	Quarterly in-app survey

02 Scope of Work

2.1 Modules In Scope

EduSphere 360 is a full-stack school management platform delivered as a web application with companion iOS and Android apps. The following modules are in scope for this engagement:

Module ID	Module Name	Priority	AI Features
M01	Student Academic Management	P0 Critical	Yes: Predictive analytics, learning gap detection
M02	Teacher KPI and Performance Management	P0 Critical	Yes: AI coaching nudges, workload balancer
M03	AI Learning Engine (Core)	P0 Critical	Yes: Fully AI-native module
M04	Attendance and Timetable System	P0 Critical	Yes: Auto-schedule, anomaly detection
M05	Communication and Parent Portal	P0 Critical	Yes: AI translation, sentiment detection
M06	Fee and Finance Management	P1 High	Yes: Payment prediction, auto-reminders
M07	Library and Resource Management	P1 High	Yes: Reading level recommender
M08	Examination and Assessment Engine	P0 Critical	Yes: AI question bank, auto-grading
M09	Extracurricular and Events Management	P1 High	Partial: Scheduling assistant
M10	Staff HR and Payroll	P1 High	Yes: Leave prediction, performance trends
M11	School Analytics and BI Dashboard	P0 Critical	Yes: Fully AI-native insights

Module ID	Module Name	Priority	AI Features
M12	Transport Management	P2 Medium	Yes: Live GPS, delay predictions
M13	Canteen and Inventory	P2 Medium	Yes: Demand forecasting
M14	Alumni and Progression Tracking	P2 Medium	Partial

2.2 Out of Scope for Version 1.0

NOTE	Native AR/VR classroom features (scheduled for v2.0) Full LMS with video hosting (integration only in v1.0) Custom curriculum builder (content import only in v1.0) Integration with government UDISE+ portal (post-launch phase) Multi-school district-level management (EduSphere 360 Enterprise edition) Blockchain-based certificate issuance (v2.0 roadmap item) Hardware integrations for biometric devices and RFID (API-ready hooks provided, hardware not in scope)
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03 User Roles and Personas

The platform serves six distinct user types. Each has fundamentally different needs, workflows, and emotional states when they arrive at the system. All UI and UX decisions must be tested against all six personas.

3.1 User Role Matrix

Role	Primary Goal	Pain Points Today	Key Features Needed
Super Admin	Full visibility and control	No unified dashboard, data in silos	Analytics, school settings, compliance reports, financials
Principal / Vice Principal	Academic and operational oversight	Reactive rather than proactive leadership	AI insights, teacher KPIs, incident management, parent comms
Class Teacher	Student welfare and teaching	3 to 4 hours per week on paperwork	Attendance, gradebook, lesson planner, student profiles
Subject Teacher	Deliver curriculum and assess	Manual test creation, grading fatigue	Question bank, AI grader, curriculum mapping, assignments
Student (Grades 3 to 12)	Learn, track progress, submit work	No personalized feedback, unclear targets	Homework portal, AI tutor, progress tracker, peer collaboration
Parent or Guardian	Know child is safe and progressing	Informed too late, multiple apps	Live attendance, grades, fees, direct teacher chat, transport
Admin Coordinator	Run day-to-day school operations	Too many manual processes	Timetable, events, HR, transport, reports
Finance Officer	Fee collection and financial reporting	Manual reconciliation, follow-up calls	Fee portal, payment dashboards, auto-reminders, ledger

3.2 Personal Spotlight: The Teacher

PERSONA	Name: Meena Krishnan Age: 34 Class Teacher plus IGCSE Biology Teacher, Grade 9 and 10
	Technical comfort: Moderate. Uses WhatsApp and Google Forms. Avoids complex tools.
	Typical morning: Marks attendance on paper, uploads to ERP, sends WhatsApp to parents about event, marks homework in separate Excel, prepares test manually.
	What she wants: One app. One login. Done in 20 minutes, not 3 hours.
	Success for her: AI drafts her lesson plan. Attendance auto-syncs. She grades; AI explains to students. Parent messages go out automatically.

04 Core Module Specifications

Each module specification includes functional requirements, AI features, UX requirements, data requirements, edge cases, and integration dependencies. This is the level of detail the development team builds against. No assumptions. No grey areas.

Module 01 Student Academic Management

BRIEF	This is the academic heartbeat of the school. Every teacher and administrator interacts with this module daily.
	It must be fast (under 2 seconds to load), intuitive (teachable in under 30 minutes), and AI-enhanced.
	All historical academic data must be importable from Excel or CSV on Day 1.

Functional Requirements

- * Student master profile: photo, contact, emergency info, medical notes, curriculum type (IGCSE, IB, or CBSE), section, blood group, learning needs
- * Gradebook: subject-wise marks entry, weighted grade calculation, grade history, class average, bell curve visualization
- * Homework and Assignment Portal: create, assign, deadline, submission, late submission policy, AI feedback
- * Lesson Plan Builder: weekly and monthly planner, curriculum alignment, resource attachment, sub-plan generation
- * Student Progress Reports: auto-generated PDF reports, custom templates, multi-language export, parent digital signature
- * Learning Needs and IEP Module: flagging students with special educational needs, custom goal tracking, resource allocation
- * Academic Calendar Integration: sync with national and school calendar, exam schedule auto-blocking
- * Peer Comparison and Benchmarking: class rank, school rank, subject percentile (anonymous to students)

AI Features for Student Academic Management

- * Early Warning System: AI detects at-risk students 4 to 6 weeks before grade drop using attendance, engagement, and assignment submission patterns
- * Learning Gap Analyser: after each test, AI maps incorrect answers to specific syllabus gaps and generates a personalised remedial plan
- * AI Tutor Chat: students can ask subject questions; AI responds aligned to their curriculum and grade level
- * Predictive Grade Forecasting: shows projected end-of-term grade based on current trajectory; updates weekly

- * Smart Assignment Feedback: AI provides draft feedback on written assignments; teacher reviews and approves before sending
- * Study Plan Generator: AI creates personalised weekly study plans based on exam schedule, weak areas, and available study time

Feature	Priority	Complexity	AI?	Notes
Student master profile	P0	Medium	No	Must support 1500+ students from day 1
Gradebook	P0	High	Partial	Weighted grades per curriculum type
Homework portal	P0	High	Yes	File upload with plagiarism check via AI
Early warning system	P0	Very High	Yes	Core differentiator; must be live from Week 1
Learning gap analyser	P1	Very High	Yes	Requires question-tag mapping in assessment module
AI tutor chat	P1	High	Yes	Curriculum-scoped; Claude or GPT-4
Progress reports PDF	P0	Medium	Partial	Must support custom school template
IEP and learning needs	P1	High	Partial	Requires separate IEP document store
Predictive grade forecast	P1	High	Yes	Minimum 6 weeks of data required to activate

Module 02 Teacher Performance and KPI Management

BRIEF	One of the most sensitive modules in the system. Teachers are professionals and must feel supported, not surveilled.
	Every KPI shown to management must also be visible to the teacher first. No surprises in appraisal meetings.
	AI insights are framed as coaching opportunities, never as punitive assessments.

Functional Requirements

- * Teacher Profile: qualifications, certifications, subjects taught, years of experience, professional development log
- * KPI Dashboard (Teacher View): personal metrics including attendance punctuality, assignment turnaround time, student satisfaction, and grade distribution
- * KPI Dashboard (Admin View): aggregated teacher performance across classes, subjects, sections, and year-on-year trends
- * Lesson Delivery Tracker: lesson plans marked complete versus planned, curriculum coverage percentage
- * Professional Development Module: training completed, certifications earned, CPD hours, external workshops
- * Observation and Appraisal Module: digital classroom observation form, 360-degree feedback from peers and principal, annual appraisal workflow

- * Workload Management: class hours per week, assignment correction load, meeting attendance, overload alerts
- * Leave and Substitution Management: leave requests, auto-match substitute teacher, affected class notifications

AI Features for Teacher KPI

- * AI Teaching Coach: weekly private suggestions to each teacher with data-driven insights on student outcomes and recommended resources
- * Workload Balance Predictor: flags when a teacher is heading toward burnout based on hours, correction load, and meetings
- * Student Satisfaction Pulse: monthly anonymous micro-survey from students with sentiment analysis and private dashboard for the teacher
- * Curriculum Coverage AI: detects gaps between planned versus delivered lessons and recommends catch-up scheduling
- * Peer Benchmarking (Anonymised): shows teacher how their class outcomes compare to same-subject teachers in the school

Module 03 AI Learning Engine (Core)

KEY MODULE	This is the module that makes EduSphere 360 different from every other school ERP in India. It is not an add-on. It is the intelligence layer that connects all other modules. The AI engine must be trained on IGCSE, IB, and CBSE curriculum content before launch. Think of it as a tireless academic advisor available to every student 24 hours a day, 7 days a week.
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Core AI Capabilities

- * Adaptive Learning Paths: after each assessment, the AI adjusts the student learning path, accelerating strong areas and reinforcing weak ones
- * Multimodal Input Support: students can type, speak, or upload a photo of their work; AI responds appropriately
- * Curriculum-Locked Responses: all AI tutor answers are scoped to the student curriculum, grade, and subject to avoid confusion or mismatch
- * Natural Language Explanation Engine: when a student gets a question wrong, AI explains why in simple language, not just provides the correct answer
- * Practice Question Generator: AI generates unlimited practice questions at the right difficulty level for each student
- * Concept Map Visualiser: AI creates visual concept maps of topics, showing how ideas connect, to help visual learners
- * Voice-Based Learning (Phase 2): text-to-speech explanations, voice questions, accessibility-first design

AI Data Model Requirements

Data Input	Model Action	Output and Benefit
Assessment scores and error patterns	Gap analysis and difficulty mapping	Personalized remedial plan per student
Time-on-task (engagement)	Attention and fatigue detection	Adaptive pacing and suggested breaks
Assignment submission timing	Procrastination detection	Nudge to student and parent alert

Data Input	Model Action	Output and Benefit
Historical grade trends (6+ weeks)	Predictive grade modelling	Forecast and intervention trigger
Teacher lesson notes	Content quality scoring	Resource recommendations to teacher
Parent communication tone	Sentiment analysis	Flag negative sentiment for admin review
Attendance patterns	Absenteeism risk model	Early alert to class teacher and counsellor

Module 04 Attendance and Timetable System

Attendance Features

- * Daily attendance: morning and afternoon, teacher-marked or auto (future biometric hook)
- * Period-wise attendance for secondary classes (Grades 9 to 12)
- * Bulk attendance import from teacher mobile with offline mode
- * Auto-SMS and notification to parent within 15 minutes of unexcused absence
- * Monthly attendance report: student-wise, class-wise, pattern analysis
- * Attendance percentage tracker: auto flag students below 75% threshold
- * Medical leave categorisation: requires doctor note upload
- * AI anomaly detection: flags unusual attendance patterns such as always absent on Mondays

Timetable Features

- * AI-powered timetable generator: inputs include subjects, teachers, rooms, periods per week, and constraints; auto-generates conflict-free timetable in under 60 seconds
- * Room allocation management: lab bookings, AV room, library periods
- * Substitute teacher auto-assignment: when teacher is absent, AI suggests best available substitute based on subject and availability
- * Timetable change notifications: push alert to affected students, parents, and teachers
- * Holiday and exam period block-out: academic calendar sync

Module 05 Communication and Parent Portal

Parents are the school's most important external stakeholders. Every interaction they have with this module either builds or erodes trust. The bar for this module is very high.

Communication Features

- * In-app messaging: teacher to parent, teacher to student, admin to all stakeholders
- * Broadcast notifications: push, SMS, email, and WhatsApp (via API) for important announcements
- * Circular management: digital circulars with read-receipt tracking and acknowledgement-required flag
- * Parent-Teacher Meeting scheduler: slot booking, agenda pre-share, meeting notes
- * Emergency broadcast: one-tap emergency alert to all parents for drill, incident, or closure
- * Multi-language support: English, Kannada, Hindi, Tamil (AI translation for parent messages)

Parent Portal Features

- * Live attendance view: "Your child checked in at 8:14 AM" type updates
- * Grade and homework visibility: current marks, upcoming deadlines, missing submissions

- * Fee payment portal: online payment, payment history, outstanding dues, instalment plans
- * Transport tracker: live bus location on map, estimated arrival time at home stop
- * Health and medical log: nurse visit records, medical history, vaccination status
- * Photo and event gallery: curated school photos, event highlights (privacy-compliant)
- * Child learning report: weekly AI-generated summary of academic progress and recommendations

Module 06 Fee and Finance Management

Fee Management

- * Fee structure builder: grade-wise, custom fee heads for tuition, bus, activity, and uniform
- * Billing engine: auto-generate invoices at start of term, recurring monthly fees, one-time fee support
- * Online payment gateway: Razorpay plus PayU plus net banking plus UPI (Bharat QR support)
- * Auto-reminder engine: SMS, WhatsApp, and email at 7 days, 3 days, day-of, and post-due-date
- * Sibling discount and scholarship management: configurable discount rules
- * Fee defaulter reporting: age-wise receivables, escalation workflow to principal
- * Digital receipts with QR code verification, downloadable PDF

Financial Reporting

- * Daily collection report: cash, online, and total
- * Month-end closing: revenue reconciliation, outstanding report
- * Budget versus actuals dashboard (admin only)
- * AI cash flow predictor: forecasts for next month's collections based on historical payment behavior

Module 07 Library and Resource Management

- * Digital catalogue: ISBN-based import, author, genre, availability status
- * Book issue and return: barcode or QR scan, due date, fine calculation
- * E-resource library: PDF, video, SCORM content linked to curriculum topics
- * AI Reading Recommender: suggests next books based on student reading history, level, and interests
- * Inter-library loan (future): connect with Bangalore public library network
- * Overdue management: automatic reminders, fine ledger, parent notification

Module 08 Examination and Assessment Engine

This module is the academic engine room. It must handle everything from a 5-minute class quiz to a formal IGCSE mock examination with zero margin for error.

- * Assessment types: unit test, midterm, final exam, quiz, assignment, practicals, viva, project
- * Question bank: tagged by subject, topic, subtopic, difficulty using Bloom's Taxonomy levels, and curriculum standard
- * AI question generator: input topic, difficulty, and count; AI generates new questions with answer keys
- * Online exam engine: time-bound, auto-submit, randomized question order, randomized options
- * Offline exam support: marks entry by teacher, auto-calculation of grades and percentile
- * Answer sheet digitization: teacher scans or uploads answer sheet image; AI performs optical mark recognition
- * Rubric-based grading: for essays and long-form answers, digital rubric with AI-assisted grading suggestion
- * Moderation workflow: second teacher review before final grades are published
- * Grade release controls: grades visible to students and parents only after principal approval
- * Analysis reports: question-wise performance, class average, standard deviation, difficulty index, discrimination index

Module 09 Extracurricular and Events Management

- * Club and activity registry: sports, music, drama, coding, debate, community service, Model UN
- * Student enrolment in activities: self-enrolment plus teacher or counsellor assignment
- * Activity attendance tracking: separate from academic attendance
- * Achievements and awards log: inter-school competitions, certificates, medals, linked to student profile
- * Event management: annual day, sports day, science fair, field trips with planning, budget, and permission slips
- * Digital permission slip: parent approves via app, no paper forms
- * Event photo management: upload, approval, privacy-safe sharing with parents
- * AI activity recommender: suggests suitable extracurricular activities for each student based on interests, academic load, and past participation

Module 10 Staff HR and Payroll

- * Employee master: all staff (teaching and non-teaching) with personal info, documents, joining date, designation, department
- * Recruitment pipeline: vacancy to application to shortlist to interview to offer to onboarding
- * Document management: offer letters, contracts, ID proofs, certificates; all digital and access-controlled
- * Leave management: earned leave, sick leave, casual leave, LOP; requests, approvals, balance tracking
- * Attendance and punch: teacher clock-in and clock-out (optional biometric hook), late arrival alerts
- * Payroll engine: CTC breakdown, tax deductions (TDS, PF, ESI), salary slip generation, bank transfer file export
- * Statutory compliance: PF, ESI, PT (Professional Tax Karnataka), Form 16 generation
- * Performance review cycle: annual review workflow with self-appraisal, manager review, rating, increment tracking
- * AI attrition risk predictor: flags employees with high flight risk based on engagement signals, leave patterns, and performance trends

Module 11 School Analytics and Business Intelligence Dashboard

FLAGSHIP	This is the executive command centre of EduSphere 360. Built for principal and management to see everything in one view. Must be visually polished, mobile-responsive, and real-time. Think of the kind of dashboard a CXO at a top company would expect. No compromise on design.
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Dashboard Widgets and Reports

- * School health score: composite AI-calculated score from 0 to 100 across academics, attendance, finance, teacher performance, and parent engagement
- * Academic performance heatmap: grade by subject matrix, colour-coded by performance band
- * Attendance trends: daily, weekly, monthly; class-wise and individual; alert threshold tracking
- * Fee collection gauge: real-time collection versus target, overdue aging analysis
- * Teacher workload distribution: identify overloaded and underutilised staff
- * Parent engagement score: app login frequency, message response rate, PTM attendance rate
- * Incident and discipline tracker: type, frequency, resolution time, repeat offenders
- * Predictive enrolment forecast: AI predicts likely dropouts and new admissions based on patterns
- * Exam performance trends: rolling 3-year academic performance versus school target bands
- * Real-time alerts panel: critical flags needing immediate attention including student at risk, fee default, teacher absence, and low attendance

Module 12 Transport Management

- * Route management: define all bus routes, stops, estimated times
- * Driver and vehicle registry: driver profile, licence, vehicle PUC, insurance due date alerts
- * Student-route assignment: map each student to pickup and drop point
- * Live GPS tracking: parent app shows real-time bus location and ETA to their stop
- * Delay notifications: auto-alert to parents if bus is 10 or more minutes delayed
- * Trip log: daily trip completion record, fuel log, maintenance schedule
- * AI route optimiser: suggests optimised route adjustments based on new student registrations

Module 13 Canteen and Inventory Management

- * Daily menu management: publish weekly menu with nutritional info and allergen flags
- * Pre-order system: parents can pre-book canteen meals; reduces food waste
- * Cashless payments: integrated with student ID or RFID card (future hook) or app-based payment
- * Inventory tracking: ingredient stock, expiry tracking, supplier management
- * Wastage reporting: track daily food wastage, trigger reorder alerts
- * AI demand forecaster: predicts daily meal demand per item based on attendance, day of week, season, and exam schedule

Module 14 Alumni and Student Progression Tracking

- * Alumni registry: graduating students automatically transition to alumni profile
- * University and career tracking: where did students go? University, stream, country; manually updated or self-reported
- * Achievement wall: notable alumni achievements shared with current students for inspiration
- * Alumni engagement: newsletters, events, mentorship connections
- * Progression analytics: track school university placement rate and IB or IGCSE score trends over years

05 AI Feature Matrix

This is a complete list of every AI use case in EduSphere 360. This is what makes the platform genuinely intelligent, not just another ERP with a chatbot bolted on. Every item below is a specific, testable AI feature.

No.	AI Feature	Serves	Trigger	Output
1	Early Academic Risk Detection	Teacher and Admin	Weekly automated scan	Alert with contributing factors and suggested actions
2	Learning Gap Map (post-test)	Student and Teacher	After each assessment graded	Personalised remedial plan PDF and in-app study path
3	AI Tutor (24/7 curriculum-aligned)	Student	Student initiates chat	Step-by-step explanation, examples, practice questions
4	Smart Homework Feedback	Student and Teacher	Assignment submitted	AI draft feedback; teacher approves; sent to student
5	Study Plan Generator	Student	Exam schedule set	Weekly study planner with topic-time allocation
6	Predictive Grade Forecast	Teacher and Parent	Weekly update	Grade trend projection with confidence band

No.	AI Feature	Serves	Trigger	Output
7	AI Question Bank Generator	Teacher	Teacher inputs parameters	Ready-to-use question set with answers and Blooms tags
8	Auto-Grading (MCQ and short answer)	Teacher	Exam submitted	Instant preliminary grades; teacher reviews before publish
9	Timetable Auto-Generator	Admin	New term setup	Conflict-free timetable in under 60 seconds
10	Substitute Teacher Recommender	Admin	Teacher marks absent	Best-match substitute from available staff list
11	Attendance Anomaly Detector	Teacher and Counsellor	Daily scan	Alert on unusual patterns such as selective absence
12	Teacher Workload Balancer	Admin and HOD	Weekly scan	Overload alert and suggested redistribution
13	AI Coaching Nudge for Teachers	Teacher	Weekly private notification	Data-driven suggestion to improve student outcomes
14	Sentiment Analysis on Parent Messages	Admin	Ongoing passive scan	Dashboard flag if negative sentiment spikes in parent comms
15	Fee Payment Predictor	Finance Officer	Monthly forecast run	Prediction of likely defaults and recommended escalation list
16	AI Cash Flow Forecast	School Management	Monthly automated report	30, 60, and 90 day cash flow projection
17	Reading Level Recommender	Student and Librarian	Book returned or login	Next book suggestions at the right reading level
18	Activity Recommender	Student and Counsellor	Profile update or end of term	Top 3 extracurricular recommendations with rationale
19	Nutrition Demand Forecaster	Canteen Manager	Daily AM forecast	Expected meal count per item for the day
20	Staff Attrition Risk Predictor	HR and Principal	Monthly scan	Risk score per staff member with contributing signals
21	Parent Engagement Score	Admin	Weekly	Score, trend, and recommended engagement nudge
22	School Health Score (Composite KPI)	Principal and Management	Real-time dashboard	Score 0 to 100 with drill-down by dimension
23	Enrolment Drop-Out Predictor	Admin and Counsellor	Term-end	Probability score and recommended retention action
24	Transport Route Optimiser	Transport Coordinator	When new student added	Suggested route adjustment to reduce distance and time
25	Curriculum Coverage Tracker	HOD and Principal	Ongoing	Percentage completion versus plan; flags syllabus risk

06 Technical Architecture and Stack

These are recommended stacks based on V-ONE Automation research. The development team may propose alternatives with written justification. No changes without approval from Saran.

6.1 Recommended Technology Stack

Layer	Technology	Justification
Frontend Web	React.js (Next.js 14+)	SEO, SSR, performance, large ecosystem
Mobile App	React Native or Progressive Web App	Single codebase, iOS and Android, offline support
Backend API	Node.js (NestJS) or Python (FastAPI)	NestJS for structured ERP logic; FastAPI for AI endpoints
Database (Primary)	PostgreSQL	Relational, ACID-compliant, proven at school scale
Database (Search)	Elasticsearch	Full-text search across student, staff, and resource data
Cache Layer	Redis	Session management, real-time notification queue
AI/ML Layer	Python (FastAPI), OpenAI API or Anthropic API	LLM features; fine-tuned models for curriculum content
File Storage	AWS S3 or Cloudflare R2	Documents, images, exam papers, report cards
Real-time Events	WebSockets (Socket.io) or Pusher	Live attendance updates, parent transport tracking
SMS and WhatsApp	Twilio or Gupshup or Kaleyra	Indian carrier support, WhatsApp Business API
Email	AWS SES or SendGrid	Transactional and bulk announcements
Payment Gateway	Razorpay (primary) plus PayU	UPI, net banking, cards; Indian market optimised
Hosting	AWS Bangalore region or GCP Mumbai	Data residency in India for PDPB compliance
CI/CD	GitHub Actions plus Docker plus Kubernetes	Containerised deployment, auto-scaling
Monitoring	Datadog or Sentry	Error tracking, performance monitoring, uptime alerts

6.2 Architecture Principles

- * Microservices-ready monolith: start as modular monolith; design for service extraction when scale demands
- * API-first design: all frontend interactions via well-documented REST APIs; GraphQL for complex data queries
- * Multi-tenancy ready: architecture must support onboarding a second school in v1.5 without data leakage risk
- * Offline-first mobile: critical teacher workflows (attendance, gradebook) must work offline with sync on reconnection
- * Zero-trust security model: every API call authenticated; role-based access control at row level
- * Data isolation: AI training data is anonymised; no personally identifiable student data leaves India

07 Integrations and Third-Party APIs

Integration	Purpose	Provider	Priority
WhatsApp Business API	Parent notifications, fee reminders, absence alerts	Gupshup or Meta WABA	P0
SMS Gateway	OTP, emergency alerts, broad reach	Twilio or Kaleyra	P0
Payment Gateway	Fee collection, online payments	Razorpay plus PayU	P0
Google Workspace or Microsoft 365	SSO login, calendar sync, Meet or Teams links	OAuth 2.0	P1
Google Maps API	Transport route mapping, live GPS display	Google Maps Platform	P1
LMS Integration	Google Classroom or Moodle import and export	REST APIs	P1
Tally or QuickBooks	Finance reconciliation export	Tally Prime API	P1
NSDL or DigiLocker	Digital document verification, Aadhaar-linked IDs	Government API	P2
OpenAI or Anthropic API	All LLM-based AI features	REST API	P0
YouTube Data API	Link educational videos to topics in library	YouTube Data API v3	P2
Firebase or OneSignal	Push notifications for mobile app	Firebase Cloud Messaging	P0
Zoom API	Online PTM scheduling, virtual parent meetings	Zoom OAuth API	P2

08 Data Privacy, Security and Compliance

NON-NEGOTIABLE

Every item in this section is mandatory from Day 1 of development. Not a post-launch fix. EduSphere 360 handles data of children. This carries the highest legal and ethical responsibility.

Any security gap is a reputational and legal risk for both the school and V-ONE Automation.

8.1 Regulatory Compliance

- * India's Personal Data Protection Bill (PDPB) 2023: explicit consent, data minimisation, right to erasure
- * IT Act 2000 Section 43A: reasonable security practices for sensitive personal data
- * CBSE and ICSE Affiliation regulations: student data retention requirements, report card archival
- * RBI Payment Aggregator guidelines: applicable to fee payment gateway implementation

8.2 Security Architecture

- * Data encryption at rest: AES-256 for all stored PII including student records, medical data, and financial information
- * Encryption in transit: TLS 1.3 minimum for all API communications
- * Row-level security: PostgreSQL RLS policies; a class teacher can only see their class students

- * JWT-based authentication with refresh token rotation; session timeout after 30 minutes of inactivity
- * Two-factor authentication: mandatory for Super Admin, Principal, and Finance Officer; optional for others
- * Audit trail: every data modification logged with user ID, timestamp, IP address, action type; immutable
- * Penetration testing: required before launch and annually thereafter
- * VAPT report: required before go-live (Vulnerability Assessment and Penetration Testing)

8.3 Child Data Protection

- * Student photos: stored with access controls; never indexed by search engines; no public URLs
- * AI tutor conversations: retained for 90 days only; not used for commercial model training
- * Parental consent: digital consent form required before student profile activation
- * Right to be forgotten: admin can permanently delete a student record on request
- * Age-appropriate UX: student-facing UI must be safe, no advertising, no data monetisation

09 Project Timeline and Delivery Phases

This is the agreed delivery schedule. Each phase has a clear scope, acceptance criteria, and payment milestone tied to it. The development team is expected to deliver working software at each phase gate.

Phase	Duration	Priority	Key Deliverables
Phase 0: Setup	Weeks 1 to 2	Critical	Tech stack confirmed, cloud infra provisioned, dev and staging environments live, database schema v1 complete, Figma design system approved
Phase 1: Core Platform	Weeks 3 to 10	Critical	Auth and RBAC, student master profiles, attendance (basic), teacher profiles, academic calendar, parent portal (read-only), admin dashboard skeleton
Phase 2: Academic Engine	Weeks 11 to 18	Critical	Gradebook, homework portal, exam engine (offline and online), fee management and payment gateway, communication module, basic AI (early warning and risk detection)
Phase 3: AI and Analytics	Weeks 19 to 26	High	Full AI learning engine, AI tutor chat, timetable auto-generator, teacher KPI dashboard, library module, transport (GPS tracking), advanced analytics dashboard
Phase 4: Operations	Weeks 27 to 32	High	HR and Payroll, canteen module, extracurricular management, alumni module, WhatsApp and SMS integration, multi-language support
Phase 5: Launch	Weeks 33 to 36	Critical	UAT with school staff, performance testing (500 concurrent users), security audit and VAPT, bug fixes, data migration, staff training and go-live

9.1 Key Milestones

Milestone	Week	Sign-off Required From
Architecture and Design Review	Week 2	Saran (V-ONE) and School IT Head

Milestone	Week	Sign-off Required From
Phase 1 Demo Day	Week 10	Saran and School Principal
Phase 2 Demo Day	Week 18	Saran and School Management
AI Engine Beta Test (with 50 students)	Week 22	Saran and Head of Academics
Full UAT Completion	Week 35	Saran, School Management, and all Department Heads
Security Clearance (VAPT Report)	Week 34	Saran and External Auditor
Production Go-Live	Week 36	Saran and School Management
30-Day Post-Launch Review	Week 40	Saran, Development Team, and School

10 Acceptance Criteria and Launch KPIs

The platform will not be accepted for production go-live unless ALL of the following criteria are met. These are binary pass-or-fail gates.

Category	Acceptance Criterion	How to Measure
Performance	Page load time under 2 seconds on 4G mobile (India average)	Chrome Lighthouse score above 85
Performance	Support 500 concurrent users without degradation	Load test report using JMeter or k6
Reliability	System uptime above 99.5% in first 30 days post-launch	Monitoring dashboard (Datadog)
Data Integrity	Zero data loss during migration from legacy system	Reconciliation report signed off by school
Security	Pass VAPT with no Critical or High severity vulnerabilities open	External auditor VAPT report
Usability	85%+ of teachers complete core tasks without training beyond 2-hour session	UAT task completion test
AI Accuracy	Early warning system achieves above 80% precision on test data set	Confusion matrix from AI team
AI Accuracy	Question generator produces curriculum-aligned output 90%+ of time	QA sample test of 100 questions
Compliance	All data stored in India (AWS Mumbai or Bangalore region)	AWS configuration audit
Mobile	iOS and Android app functional for attendance, grades, parent portal, and fee payment	QA sign-off on test devices
Integration	Razorpay live transactions successful with under 0.5% failure rate	Payment gateway test report
Accessibility	WCAG 2.1 AA compliance for student and parent-facing interfaces	Automated and manual audit

11 Assumptions, Risks and Open Questions

11.1 Assumptions

- * The school will provide a dedicated IT point of contact throughout the project
- * Existing student and staff data will be provided in structured Excel format within Week 1
- * The school has an active internet connection with minimum 500 Mbps bandwidth for the server room
- * WhatsApp Business API approval will be initiated by the school; lead time is 2 to 4 weeks
- * Third-party hardware (biometric, RFID) is not in scope for v1.0; API hooks will be provided
- * OpenAI or Anthropic API costs are a pass-through to the school; estimated INR 15,000 to 40,000 per month at full utilization

11.2 Risk Register

Risk	Likelihood	Impact	Mitigation
WhatsApp API approval delayed	High	Medium	Build SMS as fallback from Day 1; WhatsApp as enhancement
AI model hallucination in tutor module	Medium	High	Curriculum-locked prompts, human teacher review layer, student feedback flag
Legacy data quality issues (dirty Excel files)	High	High	Data audit in Week 1, cleaning tool provided, parallel-run period
Staff adoption resistance	Medium	High	Involve 5 champion teachers in UAT; gamify onboarding; Saran-led training sessions
Razorpay integration delays for school entity	Medium	Medium	Apply for Razorpay credentials in Week 2; manual fee entry as fallback
Scope creep from school management	High	High	Strict change control: all new requests go to Phase 2 backlog; signed scope document
Data breach or security incident	Low	Critical	Full security review, VAPT, ISO 27001-aligned practices from Day 1
GPS tracking accuracy (transport)	Medium	Low	Use Google Maps Platform; test on 3 routes before launch

11.3 Open Questions for Development Team

OPEN	Q1: What is your recommended approach for AI curriculum scoping: fine-tuned model, RAG, or system-prompt constraints? Please provide a technical brief by Week 2. Q2: Can your team support Kannada script in the notification system from Phase 1? Confirm Unicode and font rendering plan. Q3: For offline attendance sync, what is your conflict resolution strategy if two devices submit conflicting data? Q4: Do you have prior experience with Razorpay recurring mandate API for instalment-based fee collection? Please confirm. Q5: What is your database sharding strategy if student count grows to 5,000+ in Year 2? Q6: For the AI early warning system, are you proposing a rule-based engine, an ML model, or a hybrid? Show a prototype by Phase 1 demo.
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12 Appendix, Glossary and Notes from Saran

12.1 Glossary of Terms

Term	Definition
PRD	Product Requirements Document (this document)
P0 / P1 / P2	Priority levels: P0 = must-have for launch; P1 = important for v1.1; P2 = nice-to-have
IGCSE	International General Certificate of Secondary Education (Cambridge curriculum)
IB	International Baccalaureate (Diploma Programme, Primary Years Programme)
CBSE	Central Board of Secondary Education (Indian national curriculum)
KPI	Key Performance Indicator
UAT	User Acceptance Testing; sign-off before production release
VAPT	Vulnerability Assessment and Penetration Testing
RLS	Row-Level Security (PostgreSQL feature for data isolation)
PDPB	Personal Data Protection Bill; India primary data privacy legislation
HOD	Head of Department
PTM	Parent-Teacher Meeting
CPD	Continuing Professional Development
IEP	Individual Education Plan (for students with special learning needs)
LMS	Learning Management System
RAG	Retrieval-Augmented Generation (AI architecture pattern)
DAU	Daily Active Users
NPS	Net Promoter Score

12.2 A Note from Saran

To the Development Team,

This document represents months of thinking, dozens of conversations with school staff, parents, students, and administrators, and a genuine desire to build something that transforms how international education works in Bangalore.

I want you to read this not just as a technical specification, but as a mission brief. The children and teachers who will use EduSphere 360 deserve something that actually works; something that feels like it was built for them, not just sold to them.

If something in this document is unclear, call me. If you see a better way to solve a problem we have defined here, bring it up early. We want partners who think, not just coders who implement.

Let us build something remarkable.

Saran

Product Strategy Lead, V-ONE Automation
Bangalore, June 2026

END OF DOCUMENT

EduSphere 360 PRD v1.0 | V-ONE Automation | Prepared by Saran | Bangalore, June 2026